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# Why Does the United States Underinvest in Disaster Prevention? The Political Economy of Pre-Disaster Mitigation

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# Why Does the United States Underinvest in Disaster Prevention? The Political Economy of Pre-Disaster Mitigation

*Kingston Zhao*

**Abstract**

Despite strong evidence that every dollar invested in disaster mitigation saves about fifteen dollars in future losses, the United States still allocates the vast majority of federal disaster funding after disasters rather than before. This research investigates why post-disaster relief remains politically and institutionally prioritised over pre-disaster mitigation despite its substantially lower long-term cost and greater potential to reduce human and economic losses. It argues that U.S. disaster governance structures create political incentives that favour visible post-disaster spending while systematically underfunding preventative measures. The research aims to identify institutional reforms capable of shifting disaster finance towards mitigation while remaining politically sustainable.

Using Hurricane Katrina (2005) and the California wildfire crisis (2017–2025) as comparative case studies, the project examines how institutional design, federal legislation, insurance regulation, and electoral incentives shape disaster funding decisions. The study combines a comparative qualitative case study with financial cost-benefit analysis and policy analysis. Secondary data from FEMA, the U.S. Army Corps of Engineers, the Congressional Research Service, insurance reports, and state agencies will be analysed to compare mitigation investment with post-disaster recovery expenditure, while legislation including the Stafford Act, the Disaster Recovery Reform Act, the Building Resilient Infrastructure and Communities (BRIC) programme, and California's Proposition 103 will be evaluated to identify how existing funding mechanisms allocate resources.

## Introduction

In the United States today, there is much more money allocated for post-disaster relief because, once a president declares a disaster, the Stafford Act, a federal law that gives the president the authority to issue emergency and major disaster declarations, obligates federal payment [1]. Congress rarely fights it, making it much easier to get funding during the post-disaster relief period [2]. On the other hand, pre-disaster mitigation funding is much harder to get because it is funded through a discretionary grant program that has to be rewon in an appropriations bill every year, competing against everything else in the budget [3]. From a statistical standpoint, roughly 4% of FEMA's approved disaster funding over recent years has gone to hazard mitigation, with 68% going towards public assistance and 28% for individual housing and aid [4].

Another reason why there is less pre-disaster mitigation is that, in data from previous elections, voters have rewarded incumbents for relief spending but give no credit for preparedness. They have calculated that for almost every $1 of preparedness, it offsets roughly $15 in damage [5].

Preventing these natural disasters does not bring the government publicity, which may be a reason why the benefit of prevention is discounted. In 2018, Congress also created a dedicated pre-disaster mitigation fund, but in April of 2025, FEMA moved to end it, returning roughly $882 million of infrastructure law money. 20 states sued, and a court ruled the termination unlawful in December of 2025, and about $1 billion was reopened in March of 2026 [6]. This is a positive sign, because nobody has ever tried to cancel the disaster relief fund during any of this

time, but they have tried to cut the pre-disaster mitigation fund, even though it has proved to be more meaningful and less costly in the long run.

**Research Problem**

These facts and examples show how pre-disaster mitigation must be prioritised, especially as global warming and natural disasters continue to rise in frequency. Households usually receive an amount of money depending on the property that was damaged or lost. Hurricane Katrina and the California wildfires all show the same issues with this current setup.

This study asks why U.S. disaster funding remains concentrated after the damage occurs even though prevention costs far less, and what funding structure could shift money toward prevention in a way that survives the political pressures that have ended past attempts.

40 By comparing the response to Hurricane Katrina with the response to the California wildfires, it aims to identify a funding mechanism that reaches low-income and renter households and does not depend on annual political decisions.

**Supporting Literature and cases of natural disaster in the U.S.**

Leading scholars in the field, Raymond Burby [7] and Howard Kunreuther [8], argue that Hurricane Katrina exposed two fundamental paradoxes in U.S. disaster policy. First, federal investment in structural protection, such as levees, can unintentionally encourage development in high-risk areas, increasing future losses when protective systems fail. Second, local governments often have weak incentives to restrict development or invest in mitigation because the federal government bears much of the financial burden after disasters. Burby therefore argues that disaster policy should shift from reacting to disasters toward reducing vulnerability through comprehensive hazard planning and land-use regulation.

41 Research on the political economy of disaster relief reaches similar conclusions. Husted and Nickerson [9] find that presidential disaster declarations and federal assistance are influenced not only by disaster severity but also by electoral incentives. Presidents are more likely to approve declarations in electorally important states and where governors share the president's political party, suggesting that political considerations can shape disaster funding decisions alongside objective need.

Scholars examining the broader disaster finance system argue that the Stafford Act institutionalises this imbalance [2]. Because federal disaster relief is largely mandatory once a disaster is declared, recovery funding is politically easier to secure than mitigation funding, which competes annually through discretionary appropriations. This creates a structural bias toward ex post spending even though mitigation consistently reduces future losses and long-term public expenditure.

33 11 On August 29, 2005, Hurricane Katrina hit the Gulf Coast, and the federal levee system protecting New Orleans failed in more than 50 places, flooding almost 80% of the city, and over 1,800 people died [10]. This made the city's population drop from more than 400,000 to near zero within a week [11]. Before this tragedy, the Army Corps of Engineers repeatedly requested funding for New Orleans levee and hurricane protection work in the years immediately before

14 this disaster and was denied again and again. In fiscal year 2004, the Corps requested about $11 million, and the president's budget proposed $3 million [12]. Congress approved roughly $5.5

million. In fiscal year 2005, the Corps requested about $22.5 million and again received only $5.5 million. Overall, the construction and spending in the New Orleans district fell about 44% between 2001 and 2005, from roughly $147 million to $82 million [12]. The damage from this natural disaster was about $125 billion in 2005, which is roughly $201 billion today [13], the costliest hurricane in U.S. history. Census estimates indicate nearly half of evacuees had not returned two years later [11]. Critically, the overwhelming majority of flood damage and deaths came from the levee failure, rather than the wind or storm surge directly. The engineered system failed, not the forecast. It is estimated that roughly $50 billion in reconstruction aid and insurance payments flowed to New Orleans and its residents, about $100,000 per pre-storm resident [14]. This shows that the annual levee funding shortfall was on the order of $20 million to $30 million against the $201 billion loss [12].

California experienced a sustained escalation in wildfire destruction from the 2017-2018 fire seasons all the way through January of the 2025 Palisades and Eaton fires in Los Angeles, alongside the mass physical damage. The whole state's home insurance market began collapsing, with major insurance companies halting new policies and hundreds of thousands of homeowners being pushed onto the state's insurer as a last resort. Federal spending on this issue during this period ran about 5:1 in favor of fighting the fires over preventing them beforehand [15]. The actual cost of putting these fires out through this prolonged period was roughly $2.5 billion a year between 2016 and 2020 [16]. Fuel treatment, which is the thinning of overgrown forests and conducting controlled burns, averaged about $516 million a year between 2009 and 2018 [17]. In general, insurance is supposed to price risk, which discourages building in dangerous areas and provides money to rebuild when disaster strikes, but in this case, insurance did neither. Proposition 103, which was passed in 1988, requires insurers to get the state's approval before raising insurance rates and triggers a mandatory public hearing for any increase above 7% [18]. A loophole that insurers used was routinely requesting 6.9%, regardless of what their actual risk required. They were also barred from using forward-looking catastrophe models, meaning rates had to be based on past losses in a state where fire risk was rising fast. By 2023, seven of California's 12 largest home insurers had reduced or stopped writing these new policies [19]. The result was a market that sent no warning signal about building in fire country and then had no capacity left when the fires came. California's sustainable insurance strategy attempted a fix, permitting catastrophe models and reinsurance costs and rates in exchange for insurers writing coverage in high-risk ZIP codes, allowing more expensive properties to be insured [20]. This took effect on January 1, 2025, just a few days before the Palisades fire ignited. The Palisades and Eaton fires burned more than 37,000 acres, destroyed over 16,000 structures, and damaged 2,000 more. 29 people were killed, and both of these fires were fully contained by February 5, 2025 [21]. It is estimated at roughly $40 billion in insured losses against about $65 billion in total economic losses, making this the costliest wildfire ever recorded for the insurance industry [22]. The California Fair Plan, the state's insurer of last resort, held more than $112 billion in exposure in Los Angeles County alone as of September 2024, about 23% of its entire portfolio, and had to draw roughly $1 billion in assessments from member insurance companies to pay these claims [23]. A study of the 2018 California wildfires found that destroyed buildings accounted for 19% of the $148.5 billion in total losses, health costs from smoke exposure made up 22%, and indirect economic losses, meaning business closure, supply chain disruptions, and lost work, made up 59%, with more than half of those falling outside of California entirely [24].

**Methodology**

This study employs a mixed-methods design combining policy analysis, quantitative cost estimation, and comparative case analysis, conducted in three sequential stages. Looking at some current policies, such as the Stafford Act, which makes post-disaster payment obligatory once a president declares a disaster, while mitigation funds sit in discretionary appropriations. Next, the Disaster Recovery Reform Act of 2018 created BRIC, which is a dedicated stream. BRIC's 2025 termination and 2026 ordered reinstatement is the natural experiment in mechanism durability [6]. Proposition 103 and the sustainable insurance strategy over the state and the market side produce a classification of each funding mechanism: how exposed it is to annual political discretion.

The study will undertake a financial cost-benefit analysis comparing pre-disaster mitigation investment with post-disaster recovery expenditure using secondary data from FEMA, the U.S. Army Corps of Engineers, Congressional Research Service reports, state agencies, and insurance datasets. For both Hurricane Katrina and the California wildfires, the analysis will compare mitigation spending, realised economic losses, and recovery expenditure to assess the financial consequences of existing funding allocations. The National Institute of Building Sciences (NIBS) benefit-cost ratios will be used to estimate the avoided losses associated with increased investment in mitigation under alternative funding scenarios [25]. By comparing the economic returns to preventative and reactive spending, the analysis will identify the extent to which reallocating resources towards pre-disaster mitigation could reduce long-term public expenditure while improving resilience. Artificial intelligence will be used solely to assist with data organisation, synthesis, and scenario modelling; all analyses and policy conclusions will be independently verified using the underlying datasets and academic literature.

Comparative qualitative analysis will examine the political and institutional incentives underpinning this funding imbalance through a review of congressional reports, legislative debates, government inquiries, policy documents, academic literature, and existing research on voter behaviour and political incentives. Particular attention will be paid to how electoral incentives, public perceptions of disaster relief, media visibility, and the institutional design of the Stafford Act and related legislation shape funding decisions.

| Methodological Stage | Method | Data Sources | Expected Output |
|---|---|---|---|
| Stage 1 | Policy Analysis | Stafford Act, Disaster Recovery Reform Act (2018), BRIC, Proposition 103, California Sustainable Insurance Strategy, congressional reports, legislation | Classification of funding mechanisms and identification of institutional incentives that favour recovery over prevention. |
| Stage 2 | Financial Cost-Benefit Analysis | FEMA, U.S. Army Corps of Engineers, Congressional Research Service, National Institute of Building Sciences (NIBS), insurance reports, state agencies | Quantitative assessment of the economic efficiency of reallocating resources towards pre-disaster mitigation. |

| | | | |
|---|---|---|---|
| Stage 3 | Comparative Qualitative Case Study | Congressional hearings, government inquiries, policy documents, academic literature, media reports, voter behaviour literature | Identification of a politically durable disaster funding model capable of increasing mitigation investment while improving resilience and protecting vulnerable communities. |
| Cross-cutting Tool | Artificial Intelligence (AI) | Secondary datasets and policy documents | Improved efficiency and transparency in analysing large volumes of quantitative and qualitative data. |

**Conclusion**

This research seeks to explain why the United States continues to prioritise post-disaster relief despite substantial evidence that investment in pre-disaster mitigation is more economically efficient and socially equitable. Through a comparative analysis of Hurricane Katrina and the California wildfires, supported by financial cost-benefit analysis and policy evaluation, the study will identify the political and institutional mechanisms that shape disaster funding decisions. Rather than asking whether mitigation is beneficial, the research examines why existing governance structures continue to underinvest in prevention despite its demonstrated economic returns.

There are a few limitations. First, adding more funding to pre-disaster mitigation may incentivize locals to build in higher-risk areas [26]. Second, money cannot solve all of the problems. Pre-disaster mitigation does not mean that it will stop natural disaster damage completely; there is a high level of contingency and political and economic reasons for limited spending on disasters that may never come to pass. This will change U.S. federal budgeting from paying relief payments as natural disasters happen to a set amount every single year for pre-disaster mitigation funding that doesn't have a certain chance of stopping damage. Beyond policy and economic planning, we need to research historical disaster-cycle simulations rather than selecting one type of disaster. Next, we will also have to conduct an analysis of whether a risk index allocation formula still favors high-property-value areas like the Pacific Palisades over low-income, high-risk communities. Will also have to do a comparative case study of existing mandatory catastrophe funds, such as Florida's Hurricane Catastrophe Fund, to test transferability to a federal multi-hazard design.

U.S.'s biggest cities are not failing just because of climate change and disasters that are unpredictable. They are failing because governments are not taking action where it is needed. Pre-disaster mitigation funding has been proven to be much more sustainable and less costly in the long run. This is why action must be taken today toward a more sustainable, safer future for generations to come.

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